

Statistics from samples and Limit theorems

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Subsection 6

Concentration phenomenon

Bounding $P(|\bar{X} - \mu| > t)$

$$X_1, \dots, X_n \sim \text{iid } X$$

- Sample mean $\bar{X} = (X_1 + \dots + X_n)/n$
- Chebyshev bound

$$\mu = E[X]$$

$$\sigma^2 = \text{Var}(X)$$

$$P(|\bar{X} - \mu| > \delta) \leq \frac{\sigma^2}{n\delta^2}$$

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How weak is Chebyshev?

- Let $X \sim \text{Bernoulli}(1/2)$, $\mu = 0.5$, $\sigma^2 = 0.25$

▶ $n = 10$: $P(|\bar{X} - 0.5| > 0.3) = 0.0215 \leq 0.278$

▶ $n = 50$: $P(|\bar{X} - 0.5| > 0.3) = 5.61 \times 10^{-6} \leq 0.056$

$X_1 + \dots + X_n \sim \text{Binom}(n, 1/2)$
 $n=10$
 $P(|\bar{X} - 1/2| > 0.3)$

$= P(X_1 + \dots + X_{10} > 8 \text{ or } X_1 + \dots + X_{10} < 2)$

$\left(\frac{X_1 + \dots + X_{10} < 2}{\text{Binom}(10, 1/2)} \right)$

$n=50$
 $\text{Binom}(50, 1/2)$

$P(X_1 + \dots + X_{50} > 40 \text{ or } X_1 + \dots + X_{50} < 10)$

Chebyshev
 $\text{Binom}(10, 1/2)$

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 - ▶ $n = 50$: $P(|\bar{X} - 0.5| > 0.3) = 5.61 \times 10^{-6} \leq 0.056$
- Chebyshev falls as $1/n$
- In many cases, we can have e^{-cn}
 - ▶ Exponential fall with n
 - ▶ Much much faster than $1/n$

Markov, Chebyshev and Chernoff

- Markov inequality: X takes positive values

$$P(X > t) \leq \frac{E[X]}{t}$$

- Chebyshev inequality: X could take positive/negative values with finite variance

$$P(|X - E[X]| > t) = P((X - E[X])^2 > t^2) \leq \frac{\text{Var}(X)}{t^2}$$

Markov $\leq E[(X - E[X])^2]$

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- Chernoff inequality: X could take positive/negative values, $E[X] = 0$

$$P(X > t) = P(\overset{\text{+ve values}}{e^{\lambda X}} > e^{\lambda t}) \leq \overset{\text{markov}}{\frac{E[e^{\lambda X}]}{e^{\lambda t}}}, \quad \underline{\underline{\lambda > 0}}$$

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$$P(X > t) = P(e^{\lambda X} > e^{\lambda t}) \leq \frac{E[e^{\lambda X}]}{e^{\lambda t}}, \lambda > 0$$

- **Moment generating function (MGF)** of X : $E[e^{\lambda X}]$ (for $E[X] = 0$)
 - ▶ Pick λ that provides best bound
 - ▶ MGF too unwieldy? Use upper bound on MGF

MGF of Centralised Bernoulli(1/2)

- What is centralising?
 - ▶ X : random variable with mean $E[X]$
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 - ▶ Bernoulli(1/2): $\{0^{1/2}, 1^{1/2}\}$, mean = 1/2
 - ▶ Centralised: $X \sim \{-1/2^{1/2}, 1/2^{1/2}\}$, mean = 0

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$$E[e^{\lambda X}] = P(X = -1/2)e^{-\lambda/2} + P(X = 1/2)e^{\lambda/2} = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2}$$

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Bound on MGF

$$E[e^{\lambda X}] = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2} \leq e^{\lambda^2/4}$$

MGF and sum of *iid* random variables

$$X_1, \dots, X_n \sim \text{iid } X$$

- Let $S = X_1 + \dots + X_n$. What is MGF of S ?
- Why is this question important?
 - ▶ MGF gives bounds on $P(S > t)$
 - ▶ Sample mean $\bar{X} = S/n$

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$$E[e^{\lambda S}] = E[e^{\lambda X_1} \dots e^{\lambda X_n}] = E[e^{\lambda X_1}] \dots E[e^{\lambda X_n}] = E[e^{\lambda X}]^n$$

- **MGF of sum of independent random variables is product of the individual MGFs**

Example: Sum of centralised Benoulli

- $X \sim \text{Centralised Bernoulli}(1/2)$, i.e. $\{-1/2, 1/2\}$

$$X_1, \dots, X_n \sim \text{iid } X$$

- $S = X_1 + \dots + X_n$

$$E[e^{\lambda S}] = \left(\frac{e^{\lambda/2} + e^{-\lambda/2}}{2} \right)^n \leq e^{n\lambda^2/4}$$

- **Upper bound is so much easier than MGF!**

Chernoff bound for Binomial($n, 1/2$)

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$$\text{Chernoff} \quad P(S > t) \leq \frac{E[e^{\lambda S}]}{e^{\lambda t}} \leq e^{\underbrace{n\lambda^2/4 - \lambda t}_{\text{quadratic in } \lambda}}$$

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- Pick $\lambda = 2t/n$: $P(S > t) \leq e^{-t^2/n}$

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- Now, $Y_1 = X_1 + 1/2 \sim \text{Bernoulli}(1/2)$. So,
 $Y = Y_1 + \dots + Y_n = S + n/2 \sim \text{Binomial}(n, 1/2)$

$$P(Y > \overbrace{n/2}^{E[Y]} + \overbrace{n\delta/2}^{\delta E[Y]}) = P(S > n\delta/2) \leq e^{-n\delta^2/4}$$

$$\frac{Y}{n} - \frac{1}{2} > \frac{1}{2} \cdot \delta$$

$$t = \frac{n\delta}{2}$$

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$$P(Y > n/2 + n\delta/2) = P(S > n\delta/2) \leq e^{-n\delta^2/4}$$

- Chebyshev: $P(Y > n/2 + n\delta/2) \leq \frac{1}{n\delta^2}$

Chebyshev vs Chernoff: $Y \sim \text{Binomial}(n, 1/2)$

n	Event, $\delta = 0.6$	Prob	$1/n\delta^2$	$e^{-n\delta^2/4}$
10	$Y - 5 > 5 \times 0.6$	0.0107	0.278	0.407
50	$Y - 25 > 25 \times 0.6$	2.81×10^{-6}	0.056	0.011
100	$Y - 50 > 50 \times 0.6$	1.35×10^{-10}	0.028	1.23×10^{-4}
200	$Y - 100 > 100 \times 0.6$	4.16×10^{-19}	0.014	1.52×10^{-8}
400	$Y - 200 > 200 \times 0.6$	5.40×10^{-36}	0.007	2.32×10^{-16}

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- $1/n$ vs e^{-cn} : difference is clearly seen as n increases
 - ▶ $1/n$ is giving a very wrong idea about the magnitude of the probability

Remarks on concentration phenomenon

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- Y is the sum of the iid samples. What about other functions $f(X_1, \dots, X_n)$?
 - ▶ Many extensions: f should depend “equally” on all variables

Subsection 7

Central Limit Theorem

Moment generating function (MGF)

Definition (MGF)

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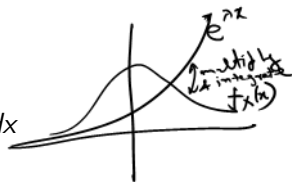
$$M_X(\lambda) = E[e^{\lambda X}].$$

- X : Discrete with PMF f_X
 - ▶ X takes values $\{x_1, x_2, \dots\}$

$$M_X(\lambda) = f_X(x_1)e^{\lambda x_1} + f_X(x_2)e^{\lambda x_2} + \dots$$

- X : continuous with PDF f_X and support T_X

$$M_X(\lambda) = \int_{x \in T_X} f_X(x) e^{\lambda x} dx$$



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- $X \sim \{-p, 1-p\}^p$ *Centralised*
 - ▶ $M_X(\lambda) = (1-p)e^{-p\lambda} + pe^{(1-p)\lambda}$

Examples

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- $X \sim \{-p^{1-p}, p\}$
 - ▶ $M_X(\lambda) = (1-p)e^{-p\lambda} + pe^{(1-p)\lambda}$
- $X \in \{-1^{1/2}, 0^{1/4}, 2^{1/4}\}$
 - ▶ $M_X(\lambda) = 0.5e^{-\lambda} + 0.25 + 0.25e^{2\lambda}$

Examples

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- $X \sim \{-p, 1-p\}$
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- $X \in \{-1, 0, 2\}$
 - ▶ $M_X(\lambda) = 0.5e^{-\lambda} + 0.25 + 0.25e^{2\lambda}$
- $M_X(\lambda) = (1/3)e^{3\lambda/2} + (1/6)e^{-3\lambda} + (1/8)e^{-\lambda} + (1/8)e^{\lambda} + 1/8$
 - ▶ $X \sim \{-3, -1, 0, 1, 3/2\}$

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- $M_X(\lambda) = \underbrace{(1/3)e^{3\lambda/2}}_{\substack{1/6 \quad 1/8 \quad 1/2 \quad 1/8 \quad 1/3 \\ \text{probabilities}}} + \underbrace{(1/6)e^{-3\lambda}}_{\substack{1/6 \quad 1/8 \quad 1/2 \quad 1/8 \quad 1/3 \\ \text{probabilities}}} + (1/8)e^{-\lambda} + (1/8)e^{\lambda} + \underbrace{1/4}_{\substack{\text{probabilities} \\ \text{at } x=0 \text{ w.p. } 1/4}} e^{\lambda \cdot 0}$
 - ▶ $X \sim \{-3, -1, 0, 1, 3/2\}$

$$E[e^{\lambda X}] = \dots + \underbrace{f_X(x_i)}_{\substack{\downarrow \\ X=x_i \text{ w.p. } f_X(x_i)}} e^{\lambda x_i} + \dots$$

$$\downarrow$$

$$X=0 \text{ w.p. } 1/4$$

- $X \sim \text{Normal}(0, \sigma^2)$

- ▶ $M_X(\lambda) = e^{\lambda^2 \sigma^2 / 2}$

Exercise

$$\int_{-\infty}^{\infty} e^{\lambda x} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} dx$$

Why Moment Generating Function?

$$E[e^{\lambda X}] = E\left[1 + \lambda X + \frac{\lambda^2}{2!}X^2 + \frac{\lambda^3}{3!}X^3 + \dots\right]$$

$$\text{mgf} = 1 + \lambda \underset{\substack{\downarrow \\ \text{1st moment}}}{E[X]} + \frac{\lambda^2}{2!} \underset{\substack{\downarrow \\ \text{2nd moment}}}{E[X^2]} + \frac{\lambda^3}{3!} \underset{\substack{\downarrow \\ \text{3rd moment}}}{E[X^3]} + \dots$$

Why Moment Generating Function?

$$\begin{aligned} E[e^{\lambda X}] &= E\left[1 + \lambda X + \frac{\lambda^2}{2!}X^2 + \frac{\lambda^3}{3!}X^3 + \dots\right] \\ &= 1 + \lambda E[X] + \frac{\lambda^2}{2!}E[X^2] + \frac{\lambda^3}{3!}E[X^3] + \dots \end{aligned}$$

- $X \sim \text{Normal}(0, \sigma^2)$, $M_X(\lambda) = e^{\lambda^2 \sigma^2 / 2}$

$$1 + \lambda E[X] + \frac{\lambda^2}{2!}E[X^2] + \frac{\lambda^3}{3!}E[X^3] + \dots = 1 + \frac{\lambda^2}{2!}\sigma^2 + \frac{\lambda^4}{4!}(3\sigma^4) + \dots$$

Handwritten note: $\frac{(\lambda^2 \sigma^2 / 2)^2}{2!}$

- $E[X] = 0$, $E[X^2] = \sigma^2$, $E[X^3] = 0$, $E[X^4] = 3\sigma^4$ and so on

Examples: Sum of two independent random variables

$$X_1, X_2 \sim \text{iid } X, Y = X_1 + X_2$$

- $X \sim \text{iid Bernoulli}(p)$ *Centralised* $\{-1, 1\}$
 - ▶ $M_X(\lambda) = (1-p)e^{-p\lambda} + pe^{(1-p)\lambda}$ *square*
 - ▶ $M_Y(\lambda) = M_X(\lambda)^2 = (1-p)^2 e^{-2p\lambda} + 2p(1-p)e^{(1-2p)\lambda} + p^2 e^{2(1-p)\lambda}$
 - ★ $Y \sim \left\{ \overset{(1-p)^2}{-2p}, \overset{2p(1-p)}{1-2p}, \overset{p^2}{2(1-p)} \right\}$ *read distribution*

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 - ▶ $M_Y(\lambda) = M_X(\lambda)^2 = (1-p)^2 e^{-2p\lambda} + 2p(1-p)e^{(1-2p)\lambda} + p^2 e^{2(1-p)\lambda}$
 - ★ $Y \sim \left\{ \overset{(1-p)^2}{-2p}, \overset{2p(1-p)}{1-2p}, \overset{p^2}{2(1-p)} \right\}$
- $X \in \left\{ \overset{1/2}{-1}, \overset{1/4}{0}, \overset{1/4}{2} \right\}$
 - ▶ $M_X(\lambda) = 0.5e^{-\lambda} + 0.25 + 0.25e^{2\lambda}$
 - ▶ $M_Y(\lambda) = 0.25e^{-2\lambda} + 0.25e^{-\lambda} + 0.0625 + 0.25e^{\lambda} + 0.125e^{2\lambda} + 0.0625e^{4\lambda}$

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- ▶ $M_X(\lambda) = (1-p)e^{-p\lambda} + pe^{(1-p)\lambda}$

- ▶ $M_Y(\lambda) = M_X(\lambda)^2 = (1-p)^2 e^{-2p\lambda} + 2p(1-p)e^{(1-2p)\lambda} + p^2 e^{2(1-p)\lambda}$

- ★ $Y \sim \left\{ \overset{(1-p)^2}{-2p}, \overset{2p(1-p)}{1-2p}, \overset{p^2}{2(1-p)} \right\}$

- $X \in \left\{ \overset{1/2}{-1}, \overset{1/4}{0}, \overset{1/4}{2} \right\}$

- ▶ $M_X(\lambda) = 0.5e^{-\lambda} + 0.25 + 0.25e^{2\lambda}$

- ▶ $M_Y(\lambda) = 0.25e^{-2\lambda} + 0.25e^{-\lambda} + 0.0625 + 0.25e^{\lambda} + 0.125e^{2\lambda} + 0.0625e^{4\lambda}$

- $M_X(\lambda) = (1/3)e^{3\lambda/2} + (1/6)e^{-3\lambda} + (1/8)e^{-\lambda} + (1/8)e^{\lambda} + 1/4$

- ▶ $M_Y(\lambda) = \frac{e^{-6t}}{36} + \frac{e^{-4t}}{24} + \frac{e^{-3t}}{12} + \frac{11e^{-2t}}{192} + \frac{1}{9}e^{-3t/2} + \frac{e^{-t}}{16} + \frac{3}{32} + \frac{e^{t/2}}{12} + \frac{e^t}{16} + \frac{1}{6}e^{3t/2} + \frac{e^{2t}}{64} + \frac{1}{12}e^{5t/2} + \frac{e^{3t}}{9}$

Example: MGF of sample mean

- Samples: $X_1, \dots, X_n \sim \text{iid } X$, $M_X(\lambda) = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2}$

▶ Sample mean: $\bar{X} = (X_1 + \dots + X_n)/n$

▶ $M_{\bar{X}/n}(\lambda) = \frac{e^{\lambda/2n} + e^{-\lambda/2n}}{2}$

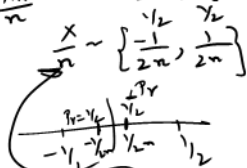
$$\frac{1}{2} e^{\lambda \cdot \frac{1}{2n}} + \frac{1}{2} e^{\lambda \cdot \frac{-1}{2n}}$$

$$M_{\bar{X}}(\lambda) = \left(\frac{e^{\frac{\lambda}{2n}} + e^{-\frac{\lambda}{2n}}}{2} \right)^n$$

fn of λ

$$\hookrightarrow \frac{X_1}{n} + \frac{X_2}{n} + \dots + \frac{X_n}{n}$$

X : Centralised
Bernoulli($\frac{1}{2}$)
 $\{-\frac{1}{2}, \frac{1}{2}\}$



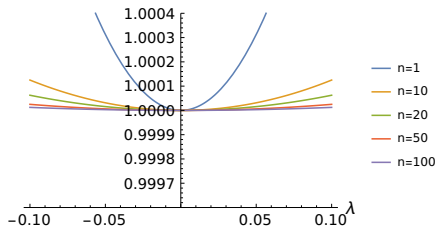
$\frac{X_1}{n}, \frac{X_2}{n}, \dots, \frac{X_n}{n} \sim \text{iid}$

Example: MGF of sample mean

- Samples: $X_1, \dots, X_n \sim \text{iid } X$, $M_X(\lambda) = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2}$
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 - ▶ $M_{\bar{X}/n}(\lambda) = \frac{e^{\lambda/2n} + e^{-\lambda/2n}}{2}$

$$M_{\bar{X}}(\lambda) = \left(\frac{e^{\frac{\lambda}{2n}} + e^{-\frac{\lambda}{2n}}}{2} \right)^n$$

MGF of sample mean for different n



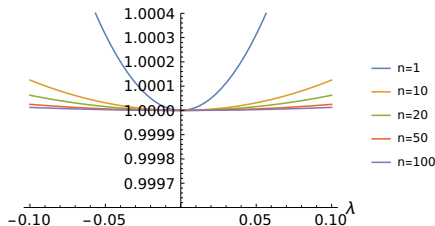
$M_{\bar{X}}(\lambda) \rightarrow 1$ as n increases

Example: MGF of sample mean

- Samples: $X_1, \dots, X_n \sim \text{iid } X$, $M_X(\lambda) = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2}$
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$$M_{\bar{X}}(\lambda) = \left(\frac{e^{\frac{\lambda}{2n}} + e^{-\frac{\lambda}{2n}}}{2} \right)^n$$

MGF of sample mean for different n



$M_{\bar{X}}(\lambda) \rightarrow 1$ as n increases

- WLLN: $\bar{X} \rightarrow E[X] = 0$
- Constant 0 has MGF = 1

MGF convergence at $1/\sqrt{n}$ scaling

- Samples: $X_1, \dots, X_n \sim \text{iid } X$, $M_X(\lambda) = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2}$
 - ▶ $E[X] = 0$, $\text{Var}(X) = 1/4$
 - ▶ Consider $Y = (X_1 + \dots + X_n)/\sqrt{n}$
 - ▶ $M_{X/\sqrt{n}}(\lambda) = \frac{e^{\lambda/2\sqrt{n}} + e^{-\lambda/2\sqrt{n}}}{2}$

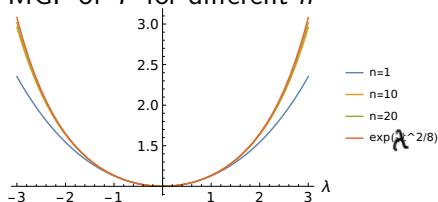
$$M_Y(\lambda) = \left(\frac{e^{\frac{\lambda}{2\sqrt{n}}} + e^{-\frac{\lambda}{2\sqrt{n}}}}{2} \right)^n$$

MGF convergence at $1/\sqrt{n}$ scaling

- Samples: $X_1, \dots, X_n \sim \text{iid } X$, $M_X(\lambda) = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2}$
 - ▶ $E[X] = 0$, $\text{Var}(X) = 1/4 \leftarrow \sigma^2$
 - ▶ Consider $Y = (X_1 + \dots + X_n)/\sqrt{n}$
 - ▶ $M_{X/\sqrt{n}}(\lambda) = \frac{e^{\lambda/2\sqrt{n}} + e^{-\lambda/2\sqrt{n}}}{2}$

$$M_Y(\lambda) = \left(\frac{e^{\frac{\lambda}{2\sqrt{n}}} + e^{-\frac{\lambda}{2\sqrt{n}}}}{2} \right)^n$$

MGF of Y for different n



$$M_Y(\lambda) \rightarrow e^{\lambda^2 \sigma^2 / 2} \text{ as } n \text{ increases}$$

$\downarrow$
Normal($0, \sigma^2$)

Another example: MGF convergence at $1/\sqrt{n}$ scaling

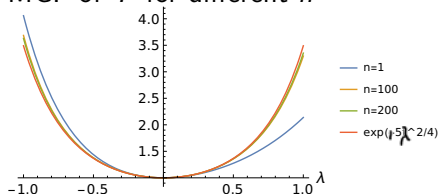
- Samples: $X_1, \dots, X_n \sim \text{iid } X$,

$$M_X(\lambda) = (1/3)e^{3\lambda/2} + (1/6)e^{-3\lambda} + (1/8)e^{-\lambda} + (1/8)e^{\lambda} + 1/4$$

- ▶ $E[X] = 0$, $\text{Var}(X) = 5/2 \sim \sigma^2$
- ▶ Consider $Y = (X_1 + \dots + X_n)/\sqrt{n}$

$$M_Y(\lambda) = \left((1/3)e^{\frac{3\lambda}{2\sqrt{n}}} + (1/6)e^{-\frac{3\lambda}{\sqrt{n}}} + (1/8)e^{-\frac{\lambda}{\sqrt{n}}} + (1/8)e^{\frac{\lambda}{\sqrt{n}}} + 1/4 \right)^n$$

MGF of Y for different n



$$M_{\bar{X}}(\lambda) \rightarrow e^{\lambda^2 \sigma^2 / 2} \text{ as } n \text{ increases}$$

$\downarrow$
 $\text{Normal}(0, \sigma^2)$

Central Limit Theorem (CLT)

Theorem (CLT)

Let $X_1, \dots, X_n \sim \text{iid } X$ with $E[X] = 0$, $\text{Var}(X) = \sigma^2$. Let $Y = (X_1 + \dots + X_n)/\sqrt{n}$. Then,

$$M_Y(\lambda) \rightarrow e^{\lambda^2 \sigma^2 / 2}$$

- MGF of $\text{Normal}(0, \sigma^2)$: $e^{\lambda^2 \sigma^2 / 2}$
- Y is said to converge in distribution to $\text{Normal}(0, \sigma^2)$

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Observations

- Contrast with WLLN
 - ▶ $\bar{X} = (X_1 + \dots + X_n)/n$ converges in distribution to $E[X]$.
- Sum of iid random variables tends to be normal
 - ▶ Do I like above statement? Not entirely.
 - ▶ Scaling is important: $1/n$ constant, $1/\sqrt{n}$ normal

Using CLT to approximate probability

$$X_1, \dots, X_n \overset{iid}{\sim} X$$

- Let $\mu = E[X]$, $\sigma^2 = \text{Var}(X)$
- $Y = X_1 + \dots + X_n$, $E[Y] = n\mu$
- What is $P(Y - n\mu > \delta n\mu)$?

Using CLT to approximate probability

$$X_1, \dots, X_n \overset{iid}{\sim} X$$

- Let $\mu = E[X]$, $\sigma^2 = \text{Var}(X)$
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Approximating using CLT

- $(Y - n\mu)/\sqrt{n}$: approximately Normal($0, \sigma^2$)

$$\frac{Y - n\mu}{\sqrt{n}\sigma} \approx \text{Normal}(0, 1)$$

Using CLT to approximate probability

$$X_1, \dots, X_n \sim X$$

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Approximating using CLT

- $(Y - n\mu)/\sqrt{n}$: approximately $\text{Normal}(0, \sigma^2)$

$$\frac{Y - n\mu}{\sqrt{n}\sigma} \approx \text{Normal}(0, 1)$$

- $F(z)$: CDF of $\text{Normal}(0,1)$ (known)
- $P(Y - n\mu > \delta n\mu) = P(\underbrace{\frac{Y - n\mu}{\sqrt{n}\sigma}}_{\approx \text{Normal}(0,1)} > \frac{\delta\sqrt{n}\mu}{\sigma}) \approx 1 - F(\frac{\delta\sqrt{n}\mu}{\sigma})$

Approximating Binomial($n, 1/2$)

$$X_1, \dots, X_n \sim \text{iid Bernoulli}(1/2)$$

- $\mu = \sigma = 1/2$
- $P(Y - n/2 > 0.6n/2) \approx 1 - F(0.6\sqrt{n})$

F : cdf of $\text{Normal}(0,1)$

n	Event, $\delta = 0.6$	Prob	$1 - F(\sqrt{n}\delta)$
10	$Y - 5 > 5 \times 0.6$	0.0107	0.0289
50	$Y - 25 > 25 \times 0.6$	2.81×10^{-6}	1.10×10^{-5}
100	$Y - 50 > 50 \times 0.6$	1.35×10^{-10}	9.87×10^{-10}

- Approximation is really close!
- Normal approximation is quite good for Binomial

Examples

$$X_1, \dots, X_n \sim \text{iid } X, Y = X_1 + \dots + X_n$$

- $X \sim \left\{ -3, -1, 0, 1, 3/2 \right\}$
 - ▶ $\mu = 0, \sigma^2 = 5/2$
 - ▶ CLT: $\frac{Y}{\sqrt{5n/2}} \approx \text{Normal}(0, 1)$
 - ▶ $P(Y > \delta n) = P\left(\frac{Y}{\sqrt{5n/2}} > \delta \sqrt{2n/5}\right) \approx 1 - F(\delta \sqrt{2n/5})$
 - ★ $n = 10, \delta = 1: \approx 0.0228$
 - ★ $n = 100, \delta = 1: \approx 1.27 \times 10^{-10}$

Examples

$$X_1, \dots, X_n \sim \text{iid } X, Y = X_1 + \dots + X_n$$

- $X \sim \{-3, -1, 0, 1, 3/2\}$
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 - ★ $n = 10, \delta = 1: \approx 0.0228$
 - ★ $n = 100, \delta = 1: \approx 1.27 \times 10^{-10}$
- $X \sim \text{Uniform}[-1, 1]$ (continuous)
 - ▶ $\mu = 0, \sigma^2 = 1/3$
 - ▶ CLT: $\sqrt{3}Y \approx \text{Normal}(0, 1)$
 - ▶ $P(Y > 0.1\sqrt{n}) = P(\sqrt{3}Y > 0.1\sqrt{3n}) \approx 1 - F(0.1\sqrt{3n})$
 - ★ $n = 10: \approx 0.2919$
 - ★ $n = 100: \approx 0.0416$

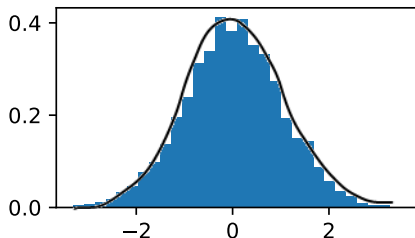
Subsection 8

Distributions, properties and connections

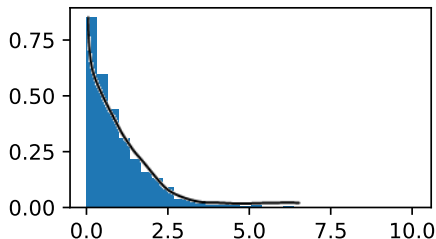
Shapes of histograms: What distribution?

shape, location,
rate, scale

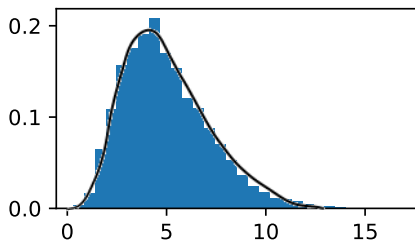
Hist1



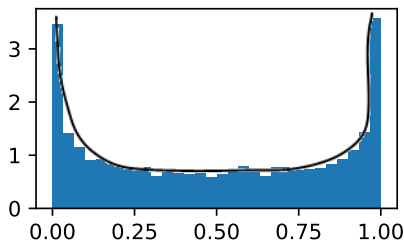
Hist2



Hist3



Hist4



Linear combination of iid normals

$$X_1, \dots, X_n \sim \overset{\text{iid}}{\text{indep}} \text{ Normal}$$

- Let $X_i \sim \text{Normal}(\mu_i, \sigma_i^2)$
- Suppose $Y = a_1X_1 + \dots + a_nX_n$
 - ▶ Linear combination of ~~iid~~ ^{indep} normals

- Then,

$$Y \sim \text{Normal}(\mu, \sigma^2)$$

where $\mu = E[Y] = a_1\mu_1 + \dots + a_n\mu_n$, $\sigma^2 = a_1^2\sigma_1^2 + \dots + a_n^2\sigma_n^2$.

- Linear combinations of ~~iid~~ ^{indep} normals is normal

- ▶ Proof: Use moment generating functions

$$\mu_1 = \dots = \mu_n = 0$$

$$E[e^{\lambda Y}] = E[e^{\lambda(a_1X_1 + \dots + a_nX_n)}] = E[e^{\lambda a_1X_1}] \dots E[e^{\lambda a_nX_n}]$$

$$e^{\frac{\lambda(a_1\sigma_1^2 + \dots + a_n\sigma_n^2)}{2}}$$

$$e^{\frac{\lambda a_1^2 \sigma_1^2}{2}} \dots e^{\frac{\lambda a_n^2 \sigma_n^2}{2}}$$

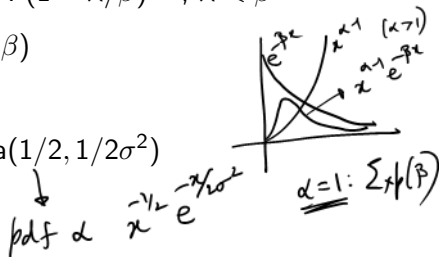
Gamma distribution

$$f_X(x) = \frac{x^{\alpha-1} e^{-\beta x}}{\int_0^{\infty} x^{\alpha-1} e^{-\beta x} dx}$$

how the distribution looks?
normal scaling

$$X \sim \text{Gamma}(\alpha, \overset{\beta}{\text{beta}}) \text{ if PDF } f_X(x) \propto \underline{x^{\alpha-1}} \underline{e^{-\beta x}}, x > 0$$

- $\alpha > 0$: shape parameter, $\beta > 0$: rate parameter, $\theta = 1/\beta$: scale parameter
- Mean: α/β , Variance: α/β^2 , MGF: $(1 - \lambda/\beta)^{-\alpha}$, $\lambda < \beta$
- Sum of n iid $\text{Exp}(\beta)$ is $\text{Gamma}(n, \beta)$
 - ▶ Proof: Use MGF (mostly)
- Square of $\text{Normal}(0, \sigma^2)$ is $\text{Gamma}(1/2, 1/2\sigma^2)$
 - ▶ Proof: Use CDF method



Cauchy distribution

$$X \sim \text{Cauchy}(\theta, \alpha^2) \text{ if PDF } f_X(x) = \frac{1}{\pi} \frac{\alpha}{\alpha^2 + (x - \theta)^2}$$

- θ : location parameter, $\alpha > 0$: scale parameter
- Mean: undefined, Variance: undefined, MGF: undefined
- Suppose $X, Y \sim \text{iid Normal}(0, \sigma^2)$. Then,

$$\frac{X}{Y} \sim \text{Cauchy}(0, 1)$$

Beta distribution

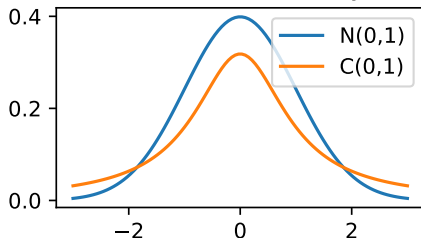
$X \sim \text{Beta}(\alpha, \beta)$ if PDF $f_X(x) \propto x^{\alpha-1}(1-x)^{\beta-1}$, $0 < x < 1$

- $\alpha > 0, \beta > 0$: shape parameters
- Mean: $\alpha/(\alpha + \beta)$, Variance: $\alpha\beta/((\alpha + \beta)^2(\alpha + \beta + 1))$
- $\text{Beta}(\alpha, 1)$ has PDF $\propto x^{\alpha-1}$: power function distribution
- Suppose $X \sim \text{Gamma}(\alpha, 1/\theta)$, $Y \sim \text{Gamma}(\beta, 1/\theta)$, then

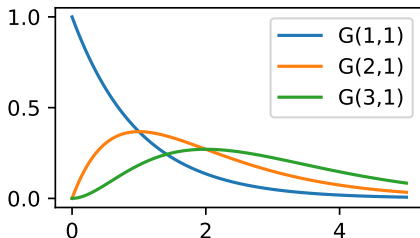
$$\frac{X}{X + Y} \overset{\text{indep}}{\sim} \text{Beta}(\alpha, \beta)$$

Plots of PDFs

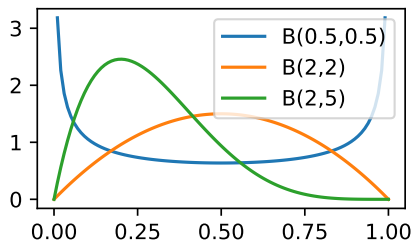
Normal and Cauchy



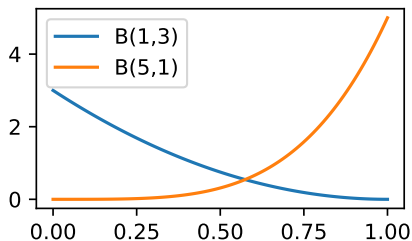
Gamma



Beta



Beta



Subsection 9

Descriptive statistics of normal samples

Normal samples

$$X_1, \dots, X_n \sim \text{iid Normal}(\mu, \sigma^2)$$

- Very common assumption in many situations
 - ▶ CLT is used as justification, often

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$$X_1, \dots, X_n \sim \text{iid Normal}(\mu, \sigma^2)$$

- Very common assumption in many situations
 - ▶ CLT is used as justification, often

- Sample mean

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

- Sample variance

$$S^2 = \frac{(X_1 - \bar{X})^2 + \dots + (X_n - \bar{X})^2}{n - 1}$$

- Recall: Sample mean and sample variance are random variables
- For normal samples, the distribution of the sample mean and variance can be characterised in more detail

Distribution of Sample Mean

$$X_1, \dots, X_n \sim \text{iid Normal}(\mu, \sigma^2)$$

- $\bar{X} = \frac{1}{n}X_1 + \dots + \frac{1}{n}X_n$
- Sample mean is a linear combination of iid normal random variables
 - ▶ So, Sample mean has a normal distribution

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- Sample mean is a linear combination of iid normal random variables
 - ▶ So, Sample mean has a normal distribution

$$\bar{X} \sim \text{Normal}(\mu, \sigma^2/n)$$

- $E[\bar{X}] = \mu$
- $\text{Var}(\bar{X}) = \frac{1}{n^2}\sigma^2 + \dots + \frac{1}{n^2}\sigma^2 = \sigma^2/n$

Sum of squares of normal samples: Chi-square

$$X_1, \dots, X_n \sim \text{iid Normal}(0, \sigma^2)$$

- X_i^2 : $\text{Gamma}(1/2, 1/2\sigma^2)$, independent
- **Result:** Sum of n independent $\text{Gamma}(\alpha, \beta)$ is $\text{Gamma}(n\alpha, \beta)$

Sum of squares of normal samples: Chi-square

$$X_1, \dots, X_n \sim \text{iid Normal}(0, \sigma^2)$$

- X_i^2 : $\text{Gamma}(1/2, 1/2\sigma^2)$, independent
- **Result:** Sum of n independent $\text{Gamma}(\alpha, \beta)$ is $\text{Gamma}(n\alpha, \beta)$

$$X_1^2 + \dots + X_n^2 \sim \text{Gamma}\left(\frac{n}{2}, \frac{1}{2\sigma^2}\right)$$

- $\text{Gamma}(n/2, 1/2)$: called Chi-square distribution with n degrees of freedom, denoted χ_n^2

Sample mean and variance of normal samples

Theorem

Suppose $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Normal}(\mu, \sigma^2)$. Then,

- 1 $\bar{X} \sim \text{Normal}(\mu, \sigma^2/n)$
- 2 $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$, Chi-square with $n - 1$ degrees of freedom.
- 3 $\bar{X}$ and S^2 are independent.

- For normal samples, the joint distribution of sample mean and variance is precisely known.

$$\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{n} \right) = \bar{X} - \bar{X} = 0$$

$$S^2 = \frac{(X_1 - \bar{X})^2 + \dots + (X_n - \bar{X})^2}{n-1}$$
$$(n-1) \frac{S^2}{\sigma^2} = \left(\frac{X_1 - \bar{X}}{\sigma} \right)^2 + \dots + \left(\frac{X_n - \bar{X}}{\sigma} \right)^2 \sim \chi_{n-1}^2$$